Assignment 10: Data Scraping

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on data scraping.

Directions

1. Rename this file `<FirstLast>_A10_DataScraping.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change "Student Name" on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up

1. Set up your session:

- Load the packages `tidyverse`, `rvest`, and any others you end up using.
- Check your working directory

```
#1

library(tidyverse); library(rvest); library(here);library(lubridate)

here()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

2. We will be scraping data from the NC DEQs Local Water Supply Planning website, specifically the Durham's 2024 Municipal Local Water Supply Plan (LWSP):

- Navigate to https://www.ncwater.org/WUDC/app/LWSP/search.php
- Scroll down and select the LWSP link next to Durham Municipality.
- Note the web address: https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-32-010&year=2024

Indicate this website as the as the URL to be scraped. (In other words, read the contents into an `rvest` webpage object.)

```r
#2

the_website <- read_html(
  'https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-32-010&year=2024'
  )
```

3. The data we want to collect are listed below:

- From the "1. System Information" section:
- Water system name
- PWSID
- Ownership
- From the "3. Water Supply Sources" section:
- Maximum Day Use (MGD) - for each month

In the code chunk below scrape these values, assigning them to four separate variables.

> HINT: The first value should be "Durham", the second "03-32-010", the third "Municipality", and the last should be a vector of 12 numeric values (represented as strings)".

```r
#3

Water_System_Name <- the_website %>%
  html_nodes('div+ table tr:nth-child(1) td:nth-child(2)') %>%
  html_text()

PWSID <- the_website %>%
  html_nodes('td tr:nth-child(1) td:nth-child(5)') %>%
  html_text()

Ownership <- the_website %>%
  html_nodes('div+ table tr:nth-child(2) td:nth-child(4)') %>%
  html_text()

Maximum_Day_Use <- the_website %>%
  html_nodes('th~ td+ td') %>%
  html_text()
```

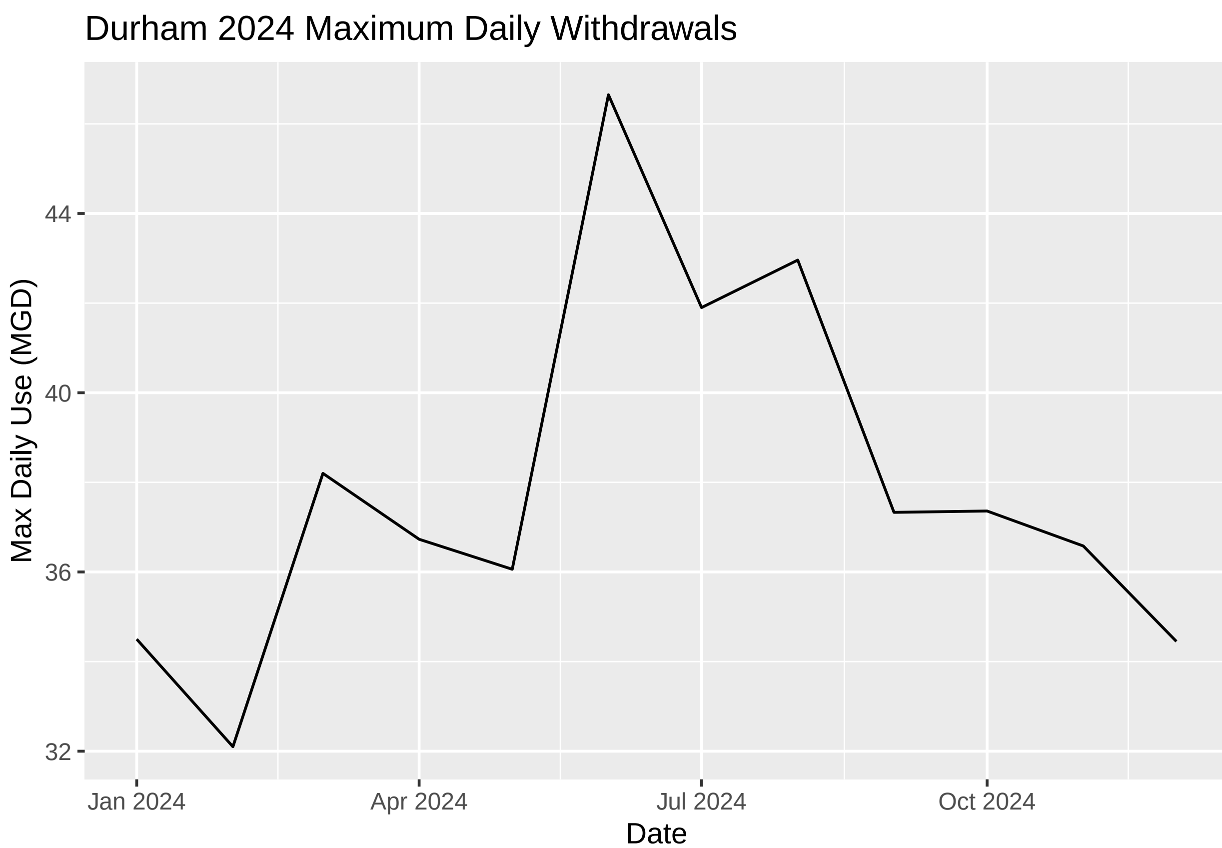
4. Convert your scraped data into a dataframe. This dataframe should have a column for each of the 4 variables scraped and a row for the month corresponding to the withdrawal data. Also add a Date column that includes your month and year in data format. (Feel free to add a Year column too, if you wish.)

   > TIP: Use `rep()` to repeat a value when creating a dataframe.

   > NOTE: It's likely you won't be able to scrape the monthly widthrawal data in chronological order. You can overcome this by creating a month column manually assigning values in the order the data are scraped: "Jan", "May", "Sept", "Feb", etc…

5. Create a line plot of the maximum daily withdrawals across the months for 2024, making sure, the months are presented in proper sequence.

```
#4

Durham2024_df <- data.frame(
  Month= (c(1,5,9,2,6,10,3,7,11,4,8,12)),
  Year= 2024,
  System.Name= Water_System_Name,
  PWSID= PWSID,
  Owner= Ownership,
  Max.Daily.Use.mgd= as.numeric(Maximum_Day_Use)
  ) %>%
  mutate(Date = my(paste(Month,"-",Year)))

#5

ggplot(data= Durham2024_df,
       aes(x= Date,y= Max.Daily.Use.mgd))+
  geom_line()+
  labs(title = "Durham 2024 Maximum Daily Withdrawals",
       y = "Max Daily Use (MGD)")
```



6. Note that the PWSID and the year appear in the web address for the page we scraped. Construct a function with two inputs - "PWSID" and "year" - that:

- Creates a URL pointing to the LWSP for that PWSID for the given year
- Creates a website object and scrapes the data from that object (just as you did above)
- Constructs a dataframe from the scraped data, mostly as you did above, but includes the PWSID and year provided as function inputs in the dataframe.
- Returns the dataframe as the function's output

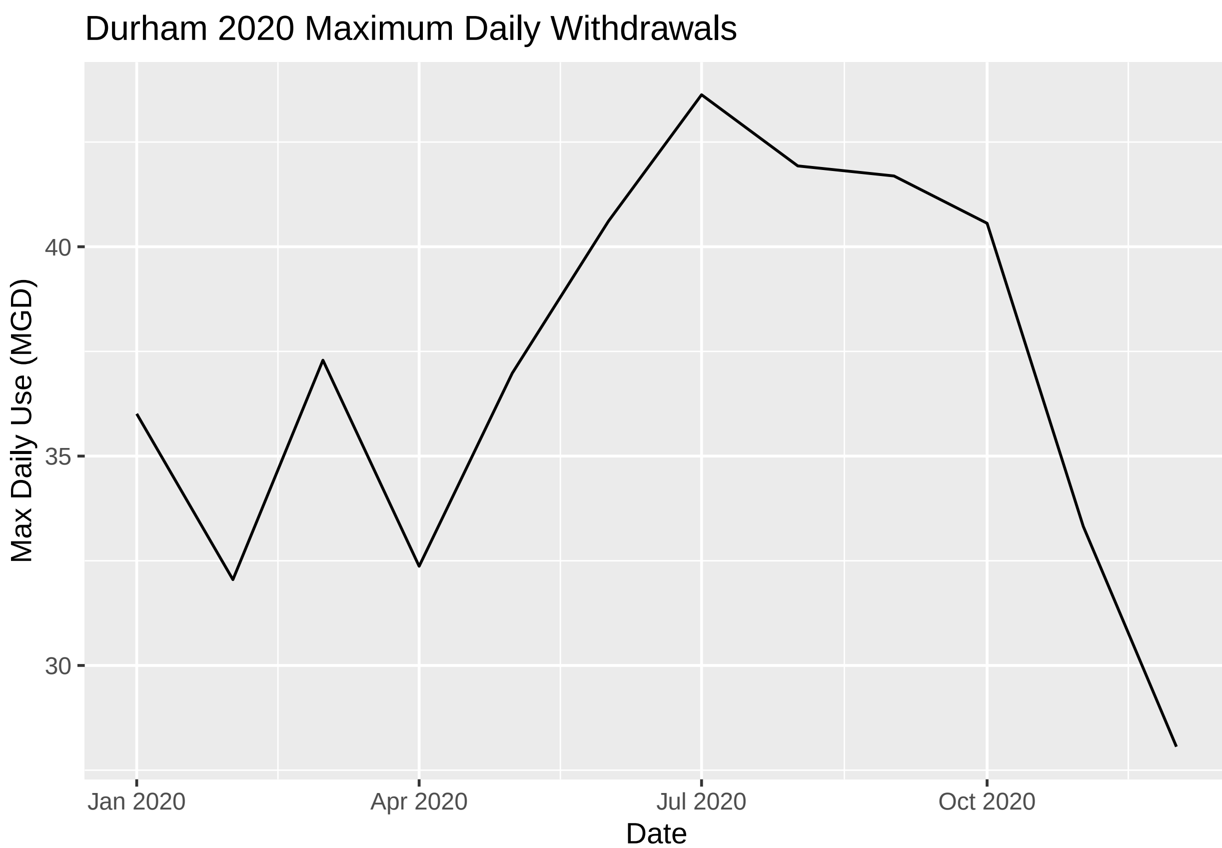
https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-32-010&year=2024

Full example web address: https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-32-010&year=2024 Base url: https://www.ncwater.org/WUDC/app/LWSP/report.php PWSID: ?pwsid=03-32-010 year: &year=2024

```
#6.

the_base_url <- 'https://www.ncwater.org/WUDC/app/LWSP/report.php'

scrape.it <- function(the_PWSID, the_Year){
  the_website <- read_html(
    paste0(the_base_url,'?pwsid=',the_PWSID,'&year=', the_Year)
    )

  Water_System_Name <-the_website %>%
    html_nodes('div+ table tr:nth-child(1) td:nth-child(2)') %>%
    html_text()

  PWSID <- the_website %>%
    html_nodes('td tr:nth-child(1) td:nth-child(5)') %>%
    html_text()

  Ownership <-the_website %>%
    html_nodes('div+ table tr:nth-child(2) td:nth-child(4)') %>%
    html_text()

  Maximum_Day_Use <-the_website %>%
    html_nodes('th~ td+ td') %>%
    html_text()

  df_MaxDailyUse<- data.frame(
    "Month"= (c(1,5,9,2,6,10,3,7,11,4,8,12)),
    "Year"= the_Year,
    "Max.Daily.Use.mgd"= as.numeric(Maximum_Day_Use),
    "PWSID" = the_PWSID) %>%

    mutate(
      System_name= Water_System_Name,
      Owner= Ownership,
      Date= my(paste(Month,"-",Year)))

  return(df_MaxDailyUse)
}
```

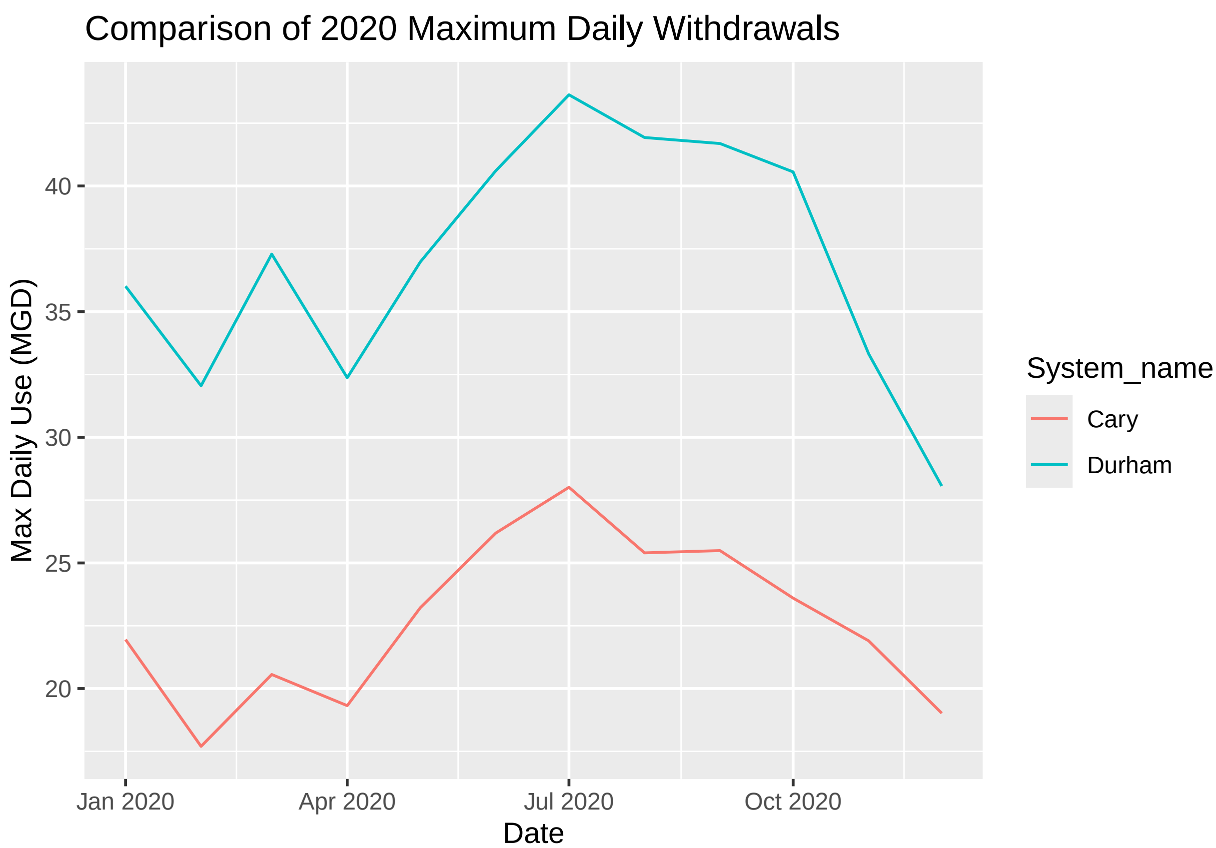
7. Use the function above to extract and plot max daily withdrawals for Durham (PWSID='03-32-010') for each month in 2020. Then show the structure of the dataframe with either the `str()` or the `glimpse()` function.

```r
#7

Durham2020_df <- scrape.it('03-32-010', 2020)

ggplot(Durham2020_df, aes(x=Date, y= Max.Daily.Use.mgd))+
  geom_line()+
  labs(title = "Durham 2020 Maximum Daily Withdrawals",
       y = "Max Daily Use (MGD)")
```



8. Use the function above to extract data for Cary (PWSID = '03-92-020') in 2020. Combine this data with the Durham data collected above and create a plot that compares Cary's to Durham's water withdrawals.

```r
#8

Cary2020_df <- scrape.it('03-92-020', 2020)

Combined_Withdrawals <- rbind(Durham2020_df, Cary2020_df)

ggplot(Combined_Withdrawals,
       aes(x= Date, y= Max.Daily.Use.mgd, color= System_name))+
  geom_line()+
  labs(title = "Comparison of 2020 Maximum Daily Withdrawals",
       y = "Max Daily Use (MGD)")
```

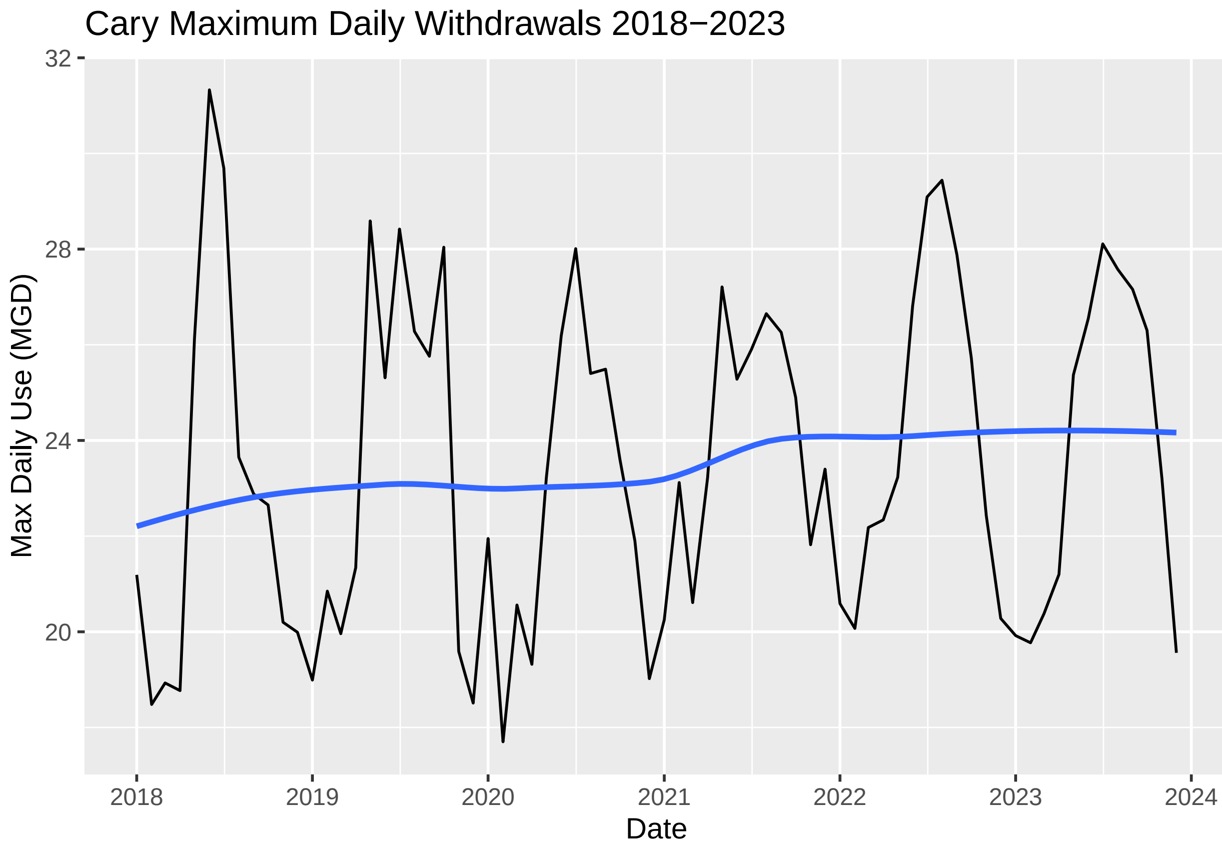


9. Use the code & function you created above to plot Cary's max daily withdrawal by months for the years 2018 thru 2023.Add a smoothed line to the plot (method = 'loess').

   TIP: See Section 3.2 in the "10_Data_Scraping.Rmd" where we apply "map2()" to iteratively run a function over two inputs. Pipe the output of the map2() function to `bind_rows()` to combine the dataframes into a single one, and use that to construct your plot.

```
#9
the_years <- as.factor(c(2018:2023))

Cary18_23 <- map2('03-92-020', the_years, scrape.it )

Cary18_23_df <- bind_rows(Cary18_23)

ggplot(Cary18_23_df,
       aes(x=Date, y=Max.Daily.Use.mgd))+
  geom_line()+
  geom_smooth(method='loess', se=FALSE)+
  scale_x_date(date_breaks = '1 year', date_labels = '%Y')+
  labs(title = "Cary Maximum Daily Withdrawals 2018-2023",
       y = "Max Daily Use (MGD)")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



Question: Just by looking at the plot (i.e. not running statistics), does Cary have a trend in water usage over time? > Answer: >Based on the plot, maximum daily withdrawals appear to have increased in Cary between 2018 and 2023.